SHACL Constraint Translation into RDF Rules

1. Path Expressions

1.1. Step 1: Rules for collecting all (sub)path-predicates

(P1) If $\langle s, path, o\rangle \in \mathscr{S}$ then $\langle o, type, Predicate\rangle \in \mathscr{R}$,

(P2) If $\langle o, type, Predicate\rangle \in \mathscr{R}$ and $o = (o_1, \ldots, o_n)$ is a list with first element o_1 and rest list $o_r = (o_2, \ldots, o_n)$ then
$\langle o_1, type, Predicate\rangle \in \mathscr{R}$
$\langle o_r, type, Predicate\rangle \in \mathscr{R}$,

(P3) If $\langle (o), type, Predicate\rangle \in \mathscr{R}$, then $\langle o, type, Predicate\rangle \in \mathscr{R}$,

(P4) If $\langle s, alternativePath, o\rangle \in \mathscr{S}$, o is a list and p is one of its elements, then
$\langle p, type, Predicate\rangle \in \mathscr{R}$,

(P5) If $\langle s, p, o\rangle \in \mathscr{S}$ with
$p \in \{inversePath, zeroOrMorePath, oneOrMorePath, zeroOrOnePath\}$ then
$\langle o, type, Predicate\rangle \in \mathscr{R}$,

1.2. Rules for collecting values after path expressions

(P6) If $\langle o, type, Predicate\rangle \in \mathscr{R}$ and $o = (o_1, \ldots, o_n)$ is a list with $n \geq 2$, o_1 is the first element and $o_r = (o_2, \ldots, o_n)$ is the rest list then
$(\langle ?x, o_1, ?y\rangle, \langle ?y, o_2, ?z\rangle) \rightarrow \langle ?x, o, ?z\rangle \in \mathscr{R}$,

(P7) If $\langle (o), type, Predicate\rangle \in \mathscr{R}$, then
$\langle ?x, o, ?y\rangle \rightarrow \langle ?x, (o), ?y\rangle \in \mathscr{R}$,

(P8) If $\langle s, alternativePath, o\rangle \in \mathscr{S}$ and o is a list and p is one of its elements then
$\langle ?x, p, ?y\rangle \rightarrow \langle ?x, o, ?y\rangle \in \mathscr{R}$,

(P9) If $\langle s, inversePath, o\rangle \in \mathscr{S}$, then
$\langle ?x, o, ?y\rangle \rightarrow \langle ?y, s, ?x\rangle \in \mathscr{R}$,

(P10) If $\langle s, zeroOrMorePath, o\rangle \in \mathscr{S}$ then
$\langle ?x, ?p, ?y\rangle \rightarrow \langle ?x, s, ?x\rangle \in \mathscr{R}$,
$\langle ?x, ?p, ?y\rangle \rightarrow \langle ?y, s, ?y\rangle, \langle ?x, o, ?y\rangle \rightarrow \langle ?x, s, ?y\rangle \in \mathscr{R}$,
$(\langle ?x, o, ?y\rangle, \langle ?y, s, ?z\rangle) \rightarrow \langle ?x, s, ?z\rangle \in \mathscr{R}$,

(P11) If $\langle s, oneOrMorePath, o\rangle \in \mathscr{S}$ then
$\langle ?x, o, ?y\rangle \rightarrow \langle ?x, s, ?y\rangle \in \mathscr{R}$,
$(\langle ?x, o, ?y\rangle, \langle ?y, s, ?z\rangle) \rightarrow \langle ?x, s, ?z\rangle \in \mathscr{R}$

(P12) If $\langle s, zeroOrOnePath, o\rangle \in \mathscr{S}$ then
$\langle ?x, ?p, ?y\rangle \rightarrow \langle ?x, s, ?x\rangle \in \mathscr{R}$,
$\langle ?x, ?p, ?y\rangle \rightarrow \langle ?y, s, ?y\rangle \in \mathscr{R}$,
$\langle ?x, o, ?y\rangle \rightarrow \langle ?x, s, ?y\rangle, \in \mathscr{R}$,

2. Focus and Value Nodes

2.1. Translation of target declarations

(T1) If $\langle s, targetnode, o\rangle \in \mathscr{S}$ then $\langle s, targets, o\rangle \in \mathscr{R}$,
(T2) If $\langle s, targetClass, o\rangle \in \mathscr{S}$ then $\langle ?x, type, o\rangle \rightarrow \langle s, targets, ?x\rangle \in \mathscr{R}$,
(T3) If $\langle s, targetSubjectOf, o\rangle \in \mathscr{S}$ then $\langle ?x, o, ?y\rangle \rightarrow \langle s, targets, ?x\rangle \in \mathscr{R}$,
(T4) If $\langle s, targetObjectOf, o\rangle \in \mathscr{S}$ then $\langle ?y, o, ?x\rangle \rightarrow \langle s, targets, ?x\rangle \in \mathscr{R}$.

2.2. Rules for obtaining value nodes

(V1) If $\langle s, path, p\rangle \in \mathscr{S}$, then $(\langle s, targets, ?x\rangle, \langle ?x, p, ?y\rangle) \rightarrow \langle s, valueNode, ?y\rangle \in \mathscr{R}$,
(V2) If $\langle s, path, p\rangle \notin \mathscr{S}$, then $\langle s, targets, ?x\rangle \rightarrow \langle s, valueNode, ?x\rangle \in \mathscr{R}$,

2.3. Rules for obtaining targets from constraints with shapes as their scope

(T5) If $\langle s, p, o\rangle \in \mathscr{R}$ with $p \in \{node, property, not, qualifiedValueShape\}$ then
$\langle s, valueNode, ?x\rangle \rightarrow \langle o, targets, ?x\rangle \in \mathscr{R}$,
(T6) If $\langle s, p, o\rangle \in \mathscr{R}$ with $p \in \{and, or, xor\}$ and o is a list and q one of its elements, then $\langle s, valueNode, ?x\rangle \rightarrow \langle q, targets, ?x\rangle \in \mathscr{R}$,

3. Constraint Checking

3.1. Rules for carrying violations (i.e. translations for node and property constraints)

(S1) If $\langle s, q, r\rangle, \langle s, path, p\rangle \in \mathscr{S}$ with $q \in \{node, property\}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle ?x, violates, r\rangle) \rightarrow \langle ?y, violates, s\rangle$,
(S2) If $\langle s, q, r\rangle \in \mathscr{S}$, $q \in \{node, property\}$, and $\langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle ?x, violates, r\rangle) \rightarrow \langle ?x, violates, s\rangle$,

3.2. Translation of Value Type Constraints

(VT1) If $\langle s, class, c\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle \_, notIncludes_{N3}, \langle ?x, type, c\rangle\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$,
(VT2) If $\langle s, class, c\rangle \in \mathscr{S}$ and $\langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle \_, notIncludes_{N3}, \langle ?x, type, c\rangle\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$,
(VT3) If $\langle s, datatype, d\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle \langle (\_?c), dtlit_{N3}, ?x\rangle,$
$\langle ?c, notEqualTo_{N3}, d\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$
(VT4) If $\langle s, datatype, d\rangle \in \mathscr{S}$ and $\langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle (\_?c), dtlit_{N3}, ?x\rangle, \langle ?c, notEqualTo_{N3}, d\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$
(VT5) If $\langle s, nodeKind_{N3}, k\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle ?x, rawType_{N3}, ?type\rangle,$
$\langle ?type, in_{N3}, t(k)\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$
(VT6) If $\langle s, nodeKind, k\rangle \in \mathscr{S}$ and $\langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle ?x, rawType_{N3}, ?type\rangle,$
$\langle ?type, in_{N3}, t(k)\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

3.3. Translation of Cardinality Constraints

(C1) If $\langle s, minCount, n\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle (?y\ (\langle ?x, p, ?y\rangle)\ ?l), collectAllIn_{N3}, \_\rangle,$
$\langle ?l, length_{N3}, ?m\rangle, \langle ?m, lessThan_{N3}, n\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

(C2) If $\langle s, maxCount, n\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle (?y\ (\langle ?x, p, ?y\rangle)\ ?l), collectAllIn_{N3}, \_\rangle,$
$\langle ?l, length_{N3}, ?m\rangle, \langle n, lessThan_{N3}, ?m\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

3.4. Translation of Value Range Constraints

(VR1) If $\langle s, minInclusive, n\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle ?x, lessThan_{N3}, n\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

(VR2) If $\langle s, minInclusive, n\rangle \in \mathscr{S}, \langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle ?x, lessThan_{N3}, n\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

(VR3) If $\langle s, minExclusive, n\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle ?x, lessThan_{N3}, n\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle ?x, equalTo_{N3}, n\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

(VR4) If $\langle s, minExclusive, n\rangle \in \mathscr{S}, \langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle ?x, lessThan_{N3}, n\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$
$(\langle s, targets, ?x\rangle, \langle ?x, equalTo_{N3}, n\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

(VR5) If $\langle s, maxInclusive, n\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle n, lessThan_{N3}, ?x\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

(VR6) If $\langle s, maxInclusive, n\rangle \in \mathscr{S}, \langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle n, lessThan_{N3}, ?x\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

(VR7) If $\langle s, maxExclusive, n\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle n, lessThan_{N3}, ?x\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle n, equalTo_{N3}, ?x\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

(VR8) If $\langle s, maxExclusive, n\rangle \in \mathscr{S}, \langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle n, lessThan_{N3}, ?x\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$
$(\langle s, targets, ?x\rangle, \langle ?x, equalTo_{N3}, n\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

3.5. Translation of String-Based Constraints

A triple with $slength_{N3}$ as its predicate will be true if the length of the string representation of the subject is in the object position. Note that $slength_{N3}$ is distinct from $length_{N3}$. In N3 syntax, both predicates have the same name, but a different prefix. A triple with $notMatches_{N3}$ as its predicate will be true if the string representation of the node in the subject position does not match the regular expression in the object position. A triple with $langTag_{N3}$ as its predicate will be true if the string in the subject position has the language tag in the object position.

(SB1) If $\langle s, minLength, n\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle ?x, slength_{N3}, ?m\rangle, \langle n, lessThan_{N3}, ?m\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

(SB2) If $\langle s, minLength, n\rangle \in \mathscr{S}, \langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle ?x, slength_{N3}, ?m\rangle, \langle n, lessThan_{N3}, ?m\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

(SB3) If $\langle s, maxLength, n\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle ?x, slength_{N3}, ?m\rangle, \langle ?m, lessThan_{N3}, n\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

(SB4) If $\langle s, maxLength, n\rangle \in \mathscr{S}, \langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle ?x, slength_{N3}, ?m\rangle, \langle ?m, lessThan_{N3}, n\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

(SB5) If $\langle s, pattern, q\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle ?x, notMatches_{N3}, q\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

(SB6) If $\langle s, pattern, q\rangle \in \mathscr{S}, \langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle ?x, notMatches_{N3}, q\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

(SB7) If $\langle s, languageIn, l\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle ?x, langTag_{N3}, ?t\rangle, \langle \_, notIncludes_{N3}, \langle ?t, in_{N3}, l\rangle\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

(SB8) If $\langle s, languageIn, l\rangle \in \mathscr{S}, \langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle ?x, langTag_{N3}, ?t\rangle, \langle \_, notIncludes_{N3}, \langle ?t, in_{N3}, l\rangle\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

(SB9) If $\langle s, uniqueLang, true\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle ?y, p, ?z\rangle, \langle \_, notIncludes_{N3}, \langle ?x, equalTo_{N3}, ?z\rangle\rangle,$
$\langle ?x, langTag_{N3}, ?t\rangle, \langle ?z, langTag_{N3}, ?u\rangle, \langle ?t, equalTo_{N3}, ?u\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

3.6. Translation of Property Pair Constraints

(PP1) If $\langle s, equals, q\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle \_, notIncludes_{N3}, \langle ?y, q, ?x\rangle\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$
$(\langle s, targets, ?y\rangle, \langle ?y, q, ?x\rangle, \langle \_, notIncludes_{N3}, \langle ?y, p, ?x\rangle\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

(PP2) If $\langle s, equals, q\rangle \in \mathscr{S}, \langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle \_, notIncludes_{N3}, \langle ?x, q, ?x\rangle\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$
$(\langle s, targets, ?x\rangle, \langle ?x, q, ?x\rangle, \langle ?x, q, ?z\rangle, \langle \_, notIncludes_{N3}, \langle ?x, equalTo_{N3}, ?z\rangle\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

(PP3) If $\langle s, disjoint, q\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle ?y, q, ?x\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

(PP4) If $\langle s, disjoint, q\rangle \in \mathscr{S}, \langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle ?x, q, ?x\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

(PP5) If $\langle s, lessThan_{N3}, q\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle ?y, q, ?z\rangle, \langle \_, notIncludes_{N3}, \langle ?x, lessThan_{N3}, ?z\rangle\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

(PP6) If $\langle s, lessThanOrEquals, q\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle ?y, q, ?z\rangle, \langle \_, notIncludes_{N3}, \langle ?x, lessThan_{N3}, ?z\rangle\rangle,$
$\langle \_, notIncludes_{N3}, \langle ?x, equalTo_{N3}, ?z\rangle\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

3.7. Translation of Logical Constraints

(L1) If $\langle s, or, l\rangle, \langle s, path, p\rangle \in \mathscr{S}$ and n is the length of list l then
$(\langle s, targets, ?z\rangle, \langle ?z, p, ?x\rangle$
$\langle (?y\ (\langle ?y, in_{N3}, l\rangle, \langle ?x, violates, ?y\rangle) ?k), collectAllIn_{N3}, \_\rangle,$
$\langle ?k, length_{N3}, ?m\rangle, \langle n, equalTo_{N3}, ?m\rangle)$
$\rightarrow \langle ?z, violates, s\rangle \in \mathscr{R}$

(L2) If $\langle s, or, l\rangle \in \mathscr{S}$, $\langle s, path, p\rangle \notin \mathscr{S}$ and n is the length of list l then
$(\langle s, targets, ?x\rangle,$
$\langle (?y\ (\langle ?y, in_{N3}, l\rangle, \langle ?x, violates, ?y\rangle) ?k), collectAllIn_{N3}, \_\rangle,$
$\langle ?k, length_{N3}, ?m\rangle, \langle n, equalTo_{N3}, ?m\rangle)$
$\rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

(L3) If $\langle s, and, l\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?z\rangle, \langle ?z, p, ?x\rangle$
$\langle (?y\ (\langle ?y, in_{N3}, l\rangle, \langle ?x, violates, ?y\rangle) ?k), collectAllIn_{N3}, \_\rangle,$
$\langle ?k, length_{N3}, ?m\rangle, \langle \_, notIncludes_{N3}, \langle 0, equalTo_{N3}, ?m\rangle\rangle)$
$\rightarrow \langle ?z, violates, s\rangle \in \mathscr{R}$

(L4) If $\langle s, and, l\rangle \in \mathscr{S}$, $\langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle,$
$\langle (?y\ (\langle ?y, in_{N3}, l\rangle, \langle ?x, violates, ?y\rangle) ?k), collectAllIn_{N3}, \_\rangle,$
$\langle ?k, length_{N3}, ?m\rangle, \langle \_, notIncludes_{N3}, \langle 0, equalTo_{N3}, ?m\rangle\rangle)$
$\rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

(L5) If $\langle s, xone, l\rangle, \langle s, path, p\rangle \in \mathscr{S}$ and n is the length of list l then
$(\langle s, targets, ?z\rangle, \langle ?z, p, ?x\rangle$
$\langle (?y\ (\langle ?y, in_{N3}, l\rangle, \langle ?x, violates, ?y\rangle) ?k), collectAllIn_{N3}, \_\rangle,$
$\langle ?k, length_{N3}, ?m\rangle, \langle n, equalTo_{N3}, ?m\rangle)$
$\rightarrow \langle ?z, violates, s\rangle \in \mathscr{R}$
$(\langle s, targets, ?z\rangle, \langle ?z, p, ?x\rangle$
$\langle (?y\ (\langle ?y, in_{N3}, l\rangle, \langle ?x, violates, ?y\rangle) ?k), collectAllIn_{N3}, \_\rangle,$
$\langle ?k, length_{N3}, ?m\rangle, \langle ?m, lessThan_{N3}, n-1\rangle)$
$\rightarrow \langle ?z, violates, s\rangle \in \mathscr{R}$

(L6) If $\langle s, xone, l\rangle \in \mathscr{S}$, $\langle s, path, p\rangle \notin \mathscr{S}$ and n is the length of list l then
$(\langle s, targets, ?x\rangle,$
$\langle (?y\ (\langle ?y, in_{N3}, l\rangle, \langle ?x, violates, ?y\rangle) ?k), collectAllIn_{N3}, \_\rangle,$
$\langle ?k, length_{N3}, ?m\rangle, \langle n, equalTo_{N3}, ?m\rangle)$
$\rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$
$(\langle s, targets, ?x\rangle,$
$\langle (?y\ (\langle ?y, in_{N3}, l\rangle, \langle ?x, violates, ?y\rangle) ?k), collectAllIn_{N3}, \_\rangle,$
$\langle ?k, length_{N3}, ?m\rangle, \langle ?m, lessThan_{N3}, n-1\rangle)$
$\rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

(L7) If $\langle s, not, b\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?z\rangle, \langle ?z, p, ?x\rangle$
$\langle \_, notIncludes_{N3}, \langle ?x, violates, b\rangle\rangle,$
$\rightarrow \langle ?z, violates, s\rangle \in \mathscr{R}$

(L8) If $\langle s, not, b\rangle \in \mathscr{S}$, $\langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?z\rangle, \langle \_, notIncludes_{N3}, \langle ?z, violates, b\rangle\rangle,$
$\rightarrow \langle ?z, violates, s\rangle \in \mathscr{R}$

3.8. Translation of QualifiedValueShape Constraint

(QVS1) If $\langle s, qualifiedValueShape, b\rangle, \langle s, qualifiedMinCount, n\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?z\rangle,$
$\langle (?y\ (\langle ?z, p, ?y\rangle) ?k), collectAllIn_{N3}, \_\rangle,$
$\langle (?v\ (\langle ?v, violates, b\rangle) ?l), collectAllIn_{N3}, \_\rangle,$
$\langle ?k, length_{N3}, ?i\rangle, \langle ?l, length_{N3}, ?j\rangle, \langle ?i - ?j, lessThan_{N3}, n\rangle)$
$\rightarrow \langle ?z, violates, s\rangle \in \mathscr{R}$

(QVS2) If $\langle s, qualifiedValueShape, b\rangle, \langle s, qualifiedMaxCount, n\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?z\rangle,$
$\langle (?y\ (\langle ?z, p, ?y\rangle)?k), collectAllIn_{N3}, \_\rangle,$
$\langle (?v\ (\langle ?v, violates, b\rangle)?l), collectAllIn_{N3}, \_\rangle,$
$\langle ?k, length_{N3}, ?i\rangle, \langle ?l, length_{N3}, ?j\rangle, \langle n, lessThan_{N3}, ?i-?j\rangle)$
$\rightarrow \langle ?z, violates, s\rangle \in \mathscr{R}$

3.9. Translation of Other Constraints

(O1) If $\langle s, in, l\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle \_, notIncludes_{N3}, \langle ?x, in, l\rangle\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

(O2) If $\langle s, in, l\rangle \in \mathscr{S}, \langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle \_, notIncludes_{N3}, \langle ?x, in, l\rangle\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

(O3) If $\langle s, hasValue, v\rangle, \langle s, path, p\rangle \in \mathscr{S}$ then
$(\langle s, targets, ?y\rangle, \langle \_, notIncludes_{N3}, \langle ?y, p, v\rangle\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

(O4) If $\langle s, hasValue, v\rangle \in \mathscr{S}, \langle s, path, p\rangle \notin \mathscr{S}$ then
$(\langle s, targets, ?x\rangle, \langle \_, notIncludes_{N3}, \langle ?x, equalTo_{N3}, v\rangle\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$

(O5) If $\langle s, closed, true\rangle, \langle s, path, p\rangle \in \mathscr{S}$ and l is the list of all paths of defined property shapes in s then
$(\langle s, targets, ?y\rangle, \langle ?y, p, ?x\rangle, \langle ?x, ?q, \_\rangle, \langle \_, notIncludes_{N3}, \langle ?q, in, l\rangle\rangle) \rightarrow \langle ?y, violates, s\rangle \in \mathscr{R}$

(O6) If $\langle s, closed, true\rangle \in \mathscr{S}, \langle s, path, p\rangle \notin \mathscr{S}$ and l is the list of all paths of defined property shapes in s then
$(\langle s, targets, ?x\rangle, \langle ?x, ?q, \_\rangle, \langle \_, notIncludes_{N3}, \langle ?q, in, l\rangle\rangle) \rightarrow \langle ?x, violates, s\rangle \in \mathscr{R}$